

Unified Binod Calendar (UBC): A Peer-Reviewed Civil Timekeeping Standard for the Binod World Government

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Abstract—This document constitutes the official specification of the Unified Binod Calendar (UBC), the civil timekeeping standard of the Republic of Binod, ratified by the Chief Scientific Committee on Temporal Standards (CSCTS). The calendar is anchored to the *Founding Epoch* of 23 April 2021 CE, 20:00 IST (Gregorian), the moment the Binodian world’s first Active Temporal Flow (ATF) session began. As of Version 2.0.0, the UBC civil date is synchronized directly to the Minecraft world: one complete Minecraft day (20 real-world minutes) equals one UBC civil day, and the calendar advances *only* during the Government’s official Active Temporal Flow window, 20:00–03:00 India Standard Time, freezing completely outside it and resuming automatically each evening. This supersedes the Version 1.0.1 model, in which the civil date was a continuous function of real elapsed Earth time via a fixed Temporal Synchronization Index (TSI), independent of Active Temporal Flow. The specification defines epoch, year structure, month nomenclature, week structure, date notation, leap corrections, and the accumulated-Minecraft-day conversion procedure. It is intended to replace all ad-hoc timekeeping practices within Binodian government, universities, and civil institutions, and to provide a robust foundation for future chronometric work.

Index Terms—civil calendar, Binod, Active Temporal Flow, UBC, chronometry, founding epoch, Minecraft day synchronization, leap correction, date conversion

Version notice (v2.0.0). This revision replaces the continuous, Temporal Synchronization Index (TSI)-based Earth-time model of Version 1.0.1 with a Minecraft-synchronized, Active Temporal Flow (ATF)-gated model: the UBC civil date is no longer a function of continuous elapsed Earth time, but of accumulated ATF-active Minecraft days since the Founding Epoch. The Founding Epoch itself is redefined from 06:00 UTC to 20:00 IST on 23 April 2021 — the exact moment the first ATF session opened — so that epoch and calendar mechanism now coincide. Calendar structure, month nomenclature, week structure, date notation, and the three-tier leap-correction scheme are **unchanged** from Version 1.0.1. Section VII and Section VIII are rewritten accordingly; TSI is retired. See Appendix B for the full changelog.

I. PREAMBLE AND SCOPE

THE Republic of Binod was founded upon the world whose first Active Temporal Flow session opened at

20:00:00 IST on 23 April 2021 CE (Gregorian proleptic). For the first years of the Republic’s existence, informal time-keeping relied upon Earth-derived Gregorian dates, creating administrative ambiguity, synchronization failures in cross-dimensional treaty instruments, and inconsistent historical records.

The CSCTS was commissioned in UBC Year 0 to design a native calendar. This document presents the ratified result. The Unified Binod Calendar:

- uses a *Founding Epoch* (FE) as its absolute zero point, coinciding with the first Active Temporal Flow (ATF) session;
- defines the civil day as one Minecraft day (20 real-world minutes), accumulated only during the Government’s official ATF window (Section II-B);
- provides a verified, auditable conversion procedure from real elapsed ATF-active time to the UBC year, month, and day;
- incorporates a periodic leap-correction mechanism sufficient for at least 10 000 UBC years without calendar drift exceeding one civil day.

This specification supersedes all prior informal timekeeping practices within Binodian jurisdiction.

II. SCIENTIFIC BACKGROUND

A. Minecraft Diurnal Cycle — Now the Civil Unit

The Binodian world operates under a local diurnal cycle of exactly 24 000 game-ticks, corresponding to 20 Earth-minutes (1 200 SI seconds) [1]. Version 1.0.1 of this standard classified this cycle as astronomical in character only, unsuitable as a civil unit because a *continuously running* Minecraft clock drifts from Earth’s calendar by approximately 70 Gregorian days per Gregorian year.

The CSCTS has since determined that this drift concern does not apply once the Minecraft clock is *gated* to a fixed, government-scheduled window rather than run continuously. Under Active Temporal Flow (ATF, Section II-B), the

Minecraft world clock advances only from 20:00 to 03:00 India Standard Time (IST) daily — a fixed, bounded 7-hour session — and is frozen for the remaining 17 hours of each Earth day. Because every civil day is tied to a fixed, auditable real-world session rather than an unbounded continuous simulation, the calendar remains exactly reproducible from the ATF schedule alone, with no historical discontinuity. The Committee therefore adopts the Minecraft day, gated by ATF, as the official civil day of the Republic of Binod, retiring the continuous Temporal Synchronization Index (TSI) model of Version 1.0.1.

B. Active Temporal Flow (ATF)

Active Temporal Flow is the Government's fixed daily schedule during which the UBC calendar — civil date and displayed civil time alike — advances:

- **Window:** 20:00 IST to 03:00 IST the following day (7 real-world hours), fixed by government policy and independent of whether the Minecraft server is running or any player is online;
- **Civil day rate:** one Minecraft day (24 000 ticks, 20 real-world minutes) equals one UBC civil day, so exactly $420/20 = 21$ civil days elapse per nightly ATF session;
- **Freeze behavior:** outside the window, the entire calendar — date, month, year, and displayed civil time — freezes exactly where it stopped and resumes automatically at the next 20:00 IST with no manual reset.

C. Civil Year Length

The civil year is designed to match the Binodian tropical year $\mathcal{Y}_B = 365.2563$ civil days (determined by meridian-transit observations recorded in CSCTS-R03 [5]), where one civil day is defined per Section II-B rather than as an absolute physical day length. This retains the same leap-correction target used since Version 1.0.0 (Section V), unaffected by the change in how a civil day accumulates.

III. EPOCH DEFINITION

- **Founding Epoch (FE):** Day 1, Month 1, Year 1 of the UBC corresponds to 23 April 2021 CE (Gregorian), 20:00:00 IST (14:30:00 UTC).
- As of Version 2.0.0, the FE is redefined from the Version 1.0.1 value of 06:00:00 UTC to 20:00:00 IST on the same Gregorian date, so that the epoch coincides exactly with the moment the first Active Temporal Flow session opened (Section II-B).
- The epoch is defined as *absolute*: it does not shift when leap corrections are applied; corrections accumulate prospectively only.
- The epoch coincides with the first verified Active Temporal Flow session of the Binodian world as established by retroactive administrative reconstruction (CSCTS-R00, Section 4.2) [4].

The **Binodian Julian Day Number** (BJDN) counts integer UBC civil days — i.e., integer accumulated Active Temporal

TABLE I
UBC MONTH NOMENCLATURE

No.	Name	Abbr.	Cultural/Astronomical Referent
1	Ardhan	ARD	Foundation; first civil sunrise
2	Ferith	FER	Iron Age; first Binodian forge
3	Selvyn	SEL	Growth; agricultural equinox
4	Kythron	KYT	Crafting; peak production cycle
5	Dravon	DRA	Trade winds; monsoon onset
6	Moriel	MOR	Mid-year zenith; Sol _B aphelion
7	Uthren	UTH	Mid-year correction; leap intercalation
8	Phiron	PHI	Descent; autumnal analog
9	Naldyn	NAL	Harvest; biome yield peak
10	Veskar	VES	Twilight; perihelion approach
11	Orrith	ORR	Astronomical winter; deep cycle
12	Caelun	CAE	Return; lengthening days
13	Zephris	ZEP	Culmination; year-end reconciliation

TABLE II
UBC WEEK DAY NAMES

Day No.	Full Name	Abbr.	Character
1	Solday	Sol	Civil work day (first)
2	Ironday	Iro	Civil work day
3	Craftday	Cra	Civil work day
4	Stonday	Sto	Civil work day
5	Woodsdays	Woo	Civil work day
6	Restday	Res	Civil rest day
7	Voidday	Voi	Civil rest / reflection day

Flow-gated Minecraft days (Section II-B) — elapsed since the FE:

$$\text{BJDN}(d = 1, m = 1, y = 1) = 1 \quad (1)$$

IV. CALENDAR STRUCTURE

A. Year

One UBC civil year contains **364 Binodian civil days** in a common year and **365 days** in a leap year (see Section V). The nominal year length of 364 days is chosen because $364 = 4 \times 7 \times 13$, enabling perfect alignment of weeks within months across all years, simplifying civil scheduling. The small divergence from the tropical year $\mathcal{Y}_B = 365.2563$ is corrected by the leap scheme.

B. Months

The year is divided into **13 months** of **28 days** each ($13 \times 28 = 364$). Each month contains exactly 4 weeks. The months are named for Binodian astronomical phenomena and cultural values as ratified by CSCTS in consultation with the Binodian Council of Heritage:

C. Week

Each month contains exactly **4 weeks of 7 days**. The seven-day week is retained for compatibility with Binodian labour-law conventions and cross-dimensional treaty obligations. Day names are:

V. LEAP CORRECTION SCHEME

A. Rationale

The tropical year $\mathcal{Y}_B = 365.2563$ civil days. A common UBC year of 364 days falls short by $\delta = 1.2563$ days per year. Without correction the calendar drifts by one full day every ≈ 0.796 years, which is unacceptable.

B. Three-Tier Correction Rules

Let y denote a UBC year number ($y \geq 1$).

Rule L1 (Annual Intercalation)

Every year a *fractional accumulator* Φ is incremented by $\delta = 1.2563$. When $\Phi \geq 1$, one intercalary day is inserted at the end of Month 7 (Uthren) and Φ is decremented by 1. Under this rule, a leap year occurs approximately every $\lfloor 1/0.2563 \rfloor = 3$ years, though in practice the interval alternates between 3 and 4 years as the fractional remainder accumulates.

Rule L2 (Century Omission)

In years that are exact multiples of 100, the L1 intercalation is *omitted* even if $\Phi \geq 1$, to correct over-accumulation of ≈ 0.2563 days over 100 years.

Rule L3 (Quad-Century Restoration)

In years that are exact multiples of 400, the L2 omission is reversed and the intercalation is performed, correcting the residual ≈ 0.0256 days per 400 years.

C. Formal Leap Test

$$\text{IsLeap}(y) = \begin{cases} \text{true} & \text{if } (y \bmod 400 = 0) \\ \text{false} & \text{else if } (y \bmod 100 = 0) \\ \text{true} & \text{else if } \Phi(y) \geq 1 \\ \text{false} & \text{otherwise} \end{cases} \quad (2)$$

where $\Phi(y) = \{(y-1) \times 1.2563\}$ (fractional part).

Long-term drift under Rules L1–L3 is less than 0.00072 days per 10000 UBC years, equivalent to ≈ 62 SI seconds — well within the precision required for civil timekeeping over historical timescales relevant to the Republic of Binod, and consistent with the periodic-correction philosophy underlying Earth’s own leap-second practice [3].

VI. DATE NOTATION STANDARD

A. Full Date Format

$$\text{DD-MMM-Y[YYY]} \quad \text{e.g., } 01\text{-ARD-1} \quad (3)$$

- DD — two-digit day within month (01–28, or 29 for intercalary day)
- MMM — three-letter month abbreviation (Table I)
- Y[YYY] — year without leading zeros

B. Numeric ISO-Compatible Format

For machine-readable interchange, following the structured, unambiguous date representation conventions of ISO 8601 [2]:

$$\text{UBCY-MM-DD} \quad \text{e.g., } \text{UBC0001-01-01} \quad (4)$$

C. Intercalary Day

When $\text{IsLeap}(y) = \text{true}$, month Uthren (07) contains 29 days. The 29th day is denoted 29-UTH-Y or UBCY-07-29 and is a mandatory public holiday designated **Synchrony Day** (*Synchara*).

VII. CONVERSION FORMULAE: ACCUMULATED ATF TIME $\rightarrow$ UBC

A. Active Temporal Flow Banking Function

Let t denote a moment in real elapsed time (UTC), and let t_{FE} denote the Founding Epoch (23 April 2021, 20:00 IST). For any calendar-day index k (counted in IST), define the nightly ATF window:

$$W(k) = [k \cdot 24\text{h} + 20\text{h}, k \cdot 24\text{h} + 27\text{h}) \quad (5)$$

(i.e., 20:00 IST on day k through 03:00 IST on day $k+1$). The **banked ATF time** $B(t)$ is the total real elapsed time between t_{FE} and t that falls inside some window $W(k)$:

$$B(t) = \sum_k |W(k) \cap [t_{\text{FE}}, t]| \quad (6)$$

$B(t)$ is monotonically non-decreasing: it increases in real time only while Active Temporal Flow is active, and is constant (frozen) at all other moments.

B. Banked Time to UBC Civil Days

One UBC civil day equals one Minecraft day, $T_{\text{civil}} = 1\,200$ s (20 real-world minutes, Section II-B). The elapsed civil days at time t are:

$$\Delta_B(t) = \frac{B(t)}{T_{\text{civil}}} = \frac{B(t)}{1\,200 \text{ s}} \quad (7)$$

C. Session-Counting Corollary

At any reference moment t that falls *between* sessions (i.e., in the frozen 17:00-hour daily interval from 03:00 to 20:00 IST), $\Delta_B(t)$ reduces to an exact integer multiple of the per-session civil-day count: if $S(t)$ complete ATF sessions have elapsed since t_{FE} , then, since each complete session contributes exactly $420 \text{ min}/20 \text{ min} = 21$ civil days (Section II-B):

$$\Delta_B(t) = 21 \cdot S(t), \quad (t \text{ between sessions}) \quad (8)$$

This corollary is a convenient special case of Eq. (7) for worked examples and audits; Eq. (6)–(7) remain the normative definition, valid continuously, including mid-session.

D. Binodian Civil Days to UBC Date

Let $N = \lfloor \Delta_B(t) \rfloor + 1$ (BJDN, 1-indexed). To find the UBC year, month, and day from N :

where:

$$\text{DaysInYear}(y) = \begin{cases} 365 & \text{if } \text{IsLeap}(y) \\ 364 & \text{otherwise} \end{cases} \quad (9)$$

Algorithm 1 BJDN to UBC Date

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1:  $y \leftarrow 1$ ;  $r \leftarrow N - 1$ 
2: while  $r \geq \text{DaysInYear}(y)$  do
3:    $r \leftarrow r - \text{DaysInYear}(y)$ 
4:    $y \leftarrow y + 1$ 
5: end while
6:  $m \leftarrow 1$ 
7: while  $r \geq \text{DaysInMonth}(y, m)$  do
8:    $r \leftarrow r - \text{DaysInMonth}(y, m)$ 
9:    $m \leftarrow m + 1$ 
10: end while
11:  $d \leftarrow r + 1$ 
12: return  $(y, m, d)$ 

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$$\text{DaysInMonth}(y, m) = \begin{cases} 29 & \text{if } m = 7 \text{ and } \text{IsLeap}(y) \\ 28 & \text{otherwise} \end{cases} \quad (10)$$

VIII. WORKED CONVERSION EXAMPLES

All examples apply Eq. (6)–(7) (Section VII), using $T_{\text{civil}} = 1\,200$ s and the Founding Epoch $t_{\text{FE}} = 23$ April 2021, 20:00 IST.

A. Example 1 — Founding Epoch: 23 April 2021, 20:00 IST

$$\begin{aligned} B(t_{\text{FE}}) &= 0 \text{ s (by definition)} \\ \Delta_B &= 0/1\,200 = 0 \\ N &= \lfloor 0 \rfloor + 1 = 1 \\ \Rightarrow & \mathbf{01-ARD-1} \quad (\text{UBC0001-01-01}) \end{aligned} \quad (11)$$

This is Solday, the first day of the Republic, and the moment the first Active Temporal Flow session opened.

B. Example 2 — 23 April 2021, 20:20 IST (20 real minutes later)

Twenty real-world minutes into the first ATF session, exactly one civil day has banked:

$$\begin{aligned} B(t) &= 1\,200 \text{ s (20 real minutes, fully within the first ATF window)} \\ \Delta_B &= 1\,200/1\,200 = 1 \\ N &= \lfloor 1 \rfloor + 1 = 2 \\ \Rightarrow & \mathbf{02-ARD-1} \quad (\text{UBC0001-01-02}) \end{aligned} \quad (12)$$

Confirms Eq. (6): every 20 real-world minutes of ATF-active time advances the civil date by exactly one day.

C. Example 3 — 2 July 2026, 05:30 IST (between sessions)

This reference moment falls in the frozen interval (03:00–20:00 IST), so the Session-Counting Corollary (Eq. (8)) applies directly. As of this moment, $S(t) = 1\,896$ complete Active Temporal Flow sessions have elapsed since the Founding Epoch:

$$\begin{aligned} \Delta_B(t) &= 21 \times 1\,896 = 39\,816 \\ N &= \lfloor 39\,816 \rfloor + 1 = 39\,817 \end{aligned} \quad (13)$$

Applying Algorithm 1: every year is common (364 days) below Year 400 (Section V), so 108 complete common years span $108 \times 364 = 39\,312$ days, leaving $r = 39\,816 - 39\,312 = 504$. Year 109 (common, 364 days) consumes 364 of these, leaving $r = 504 - 364 = 140$; since $140 < 364$, Year 110 is reached with $r = 140$, so $y = 110$. Month: $\lfloor 140/28 \rfloor = 5$ full months consumed exactly (140 days, remainder 0), so the date falls on day 1 of month 6 (MOR):

$$2 \text{ July } 2026, 05:30 \text{ IST} \rightarrow \mathbf{01-MOR-110} \quad (\text{UBC0110-06-01}) \quad (14)$$

This is Solday: 39 816 civil days since the Founding Epoch is an exact multiple of 28 (four complete weeks per month), so day 1 of any month always falls on Solday — see Section IV-C.

D. Example 4 — Mid-Session Reference

At 20:50 IST on the evening following Example 3 (30 real minutes into that night’s ATF session), one additional civil day has fully banked and a second is 1/2 elapsed:

$$\begin{aligned} \Delta_B(t) &= 39\,816 + \frac{1\,800}{1\,200} = 39\,817.5 \\ N &= \lfloor 39\,817.5 \rfloor + 1 = 39\,818 \\ \Rightarrow & \mathbf{02-MOR-110} \quad (\text{UBC0110-06-02, 50\% elapsed}) \end{aligned} \quad (15)$$

IX. IMPLEMENTATION SYNCHRONIZATION DIAGNOSTICS

Because the UBC calendar is now defined entirely by the Active Temporal Flow banking function $B(t)$ (Eq. (6)), rather than by a physical constant subject to secular drift, long-term synchronization reduces to verifying that $B(t)$ is computed correctly and continuously, rather than periodically re-measuring a coupling constant as under the retired v1.0.1 TSI model.

1) *Banking Continuity Check:* Reference implementations SHOULD record, at each observation, the banked ATF-active seconds and the timestamp of that observation. On the next observation, while ATF remains continuously active, the expected banked total is the previous value plus the real elapsed time between observations; any discrepancy (*clock drift*) indicates an implementation fault (e.g., a missed tick or a clock adjustment on the host system) rather than a property of the calendar itself, since $B(t)$ is, by Eq. (6), a deterministic function of real elapsed time and the fixed ATF schedule alone.

2) *Full Recomputation:* Because N and the in-day clock are pure functions of $B(t)$ (Eq. (7)), an implementation MAY always discard any cached or persisted state and recompute the current UBC date and time in full from the Founding Epoch. This makes the system self-healing: a page reload, application restart, or storage failure never desynchronizes the displayed calendar.

TABLE III
UBC FUNDAMENTAL CONSTANTS (VERSION 2.0.0)

Symbol	Value	Description
t_{FE}	23 Apr 2021, 20:00 IST	Founding Epoch
T_{civil}	1 200 s (20 min)	UBC civil day = 1 Minecraft day
W_{ATF}	20:00–03:00 IST	Active Temporal Flow window
L_{sess}	420 min (7 h)	Real-time length of one ATF session
n_{civil}	21 days	Civil days per ATF session
$\mathcal{Y}_B$	365.2563 civil days	Binodian tropical year (target)
δ	1.2563 day/year	Annual leap accumulation rate
D_{common}	364 days	Common UBC year length
D_{leap}	365 days	Leap UBC year length

X. CIVIL ADMINISTRATION GUIDELINES

A. Official Transition Date

All Binodian government documents issued on or after **01-ARD-5 (UBC0005-01-01)** must use UBC notation as the primary date format. Gregorian dates may appear in parentheses as secondary notation during the 10-year transition period (UBC 5–15).

B. Historical Documents

Pre-Founding-Epoch dates shall be expressed using the prefix “**pFE**” (pre-Founding Epoch) followed by the Gregorian year in parentheses. Example: pFE (2019–CE).

C. Cross-Dimensional Treaties

When Binodian dates appear in legal instruments with Earth-jurisdiction parties, both UBC and Gregorian dates must be stated. The Gregorian date shall be computed using the inverse of Eq. (7) and recorded to the nearest Earth day.

XI. SUMMARY OF CONSTANTS

XII. CONCLUSION

The Unified Binod Calendar provides an administratively practical, government-auditable timekeeping system for the Republic of Binod. By anchoring to the Founding Epoch of 23 April 2021 CE, 20:00 IST — the moment the first Active Temporal Flow session opened — and computing every date from accumulated ATF-gated Minecraft days rather than a continuous physical constant, the UBC ensures that every Binodian date corresponds to a fixed, reproducible government schedule rather than an uncontrolled real-time process. The 13-month, 28-day-per-month structure eliminates intra-year calendar irregularities, while the three-tier leap scheme guarantees sub-day accuracy over 10 000 UBC years. The Committee recommends immediate adoption by all branches of Binodian government and invites external peer review from any institution operating under a recognized civil calendar standard.

TABLE IV
SELECTED EARTH–UBC CONVERSION REFERENCE (REGENERATED FROM ALGORITHM 1, V2.0.0)

Gregorian Date	UBC Date	Significance
23 Apr 2021, 20:00 IST	01-ARD-1 (UBC0001-01-01)	Founding Epoch
01 Jan 2023, 05:30 IST*	15-NAL-36 (UBC0036-09-15)	New Gregorian Year
02 Jul 2026, 05:30 IST*	01-MOR-110 (UBC0110-06-01)	Present (near)

*Between sessions (03:00–20:00 IST); Eq. (8) applies.

APPENDIX A CONVERSION REFERENCE TABLE

APPENDIX B CHANGELOG

v1.0.0 (04-KYT-5): Initial ratified standard. Defined the Founding Epoch at 06:00 UTC, a continuous Earth-time model driven by a fixed Temporal Synchronization Index ($TSI = 86\,400/86\,736$), and the 13-month/28-day calendar structure with three-tier leap correction.

v1.0.1 (01-SEL-6): Corrected arithmetic errors in the worked Gregorian → UBC conversion examples, which had computed the Julian Date of non-epoch dates inconsistently with the Founding Epoch’s own Julian Date definition, and added a normative Julian Date convention to make the computation unambiguous. No change was made to the epoch, calendar structure, leap-correction rules, TSI value, or the conversion algorithm.

v2.0.0 (01-MOR-110): Fundamental redesign. Retired the continuous, TSI-based Earth-time model entirely. The UBC civil date is now synchronized directly to the Minecraft world and advances only during the Government’s Active Temporal Flow (ATF) window, 20:00–03:00 IST: one Minecraft day (20 real-world minutes of ATF-active time) equals one UBC civil day (Section II-B). The Founding Epoch is redefined from 06:00 UTC to 20:00 IST on the same Gregorian date (23 April 2021), coinciding with the first ATF session. Section VII (formerly Julian-Date/TSI-based) and Section VIII are rewritten around the new Active Temporal Flow Banking Function, Eq. (6). Calendar structure, month nomenclature, week structure, date notation, and the three-tier leap-correction scheme (Section V) are unchanged. Reference implementations must not reuse v1.0.1 example values, which were computed under the retired continuous model and do not apply here; validate against Table IV instead.

APPENDIX C RATIFICATION RECORD

This standard was ratified by the following CSCTS subcommittees:

- Sub-Committee on Orbital Mechanics — unanimous approval
- Sub-Committee on Civil Administration — unanimous approval
- Sub-Committee on Historical Records — unanimous approval

- Sub-Committee on Cross-Dimensional Treaty Law — approved with notation

Effective date: 04-KYT-5 (UBC0005-04-04)

Next scheduled review: 01-ARD-15 (UBC0015-01-01)

APPENDIX D

NOTE ON THE RETIRED GAMEPLAY-LAYER APPENDIX

Prior to Version 2.0.0, this appendix separately documented Active Temporal Flow (ATF) as an *informative*, non-normative gameplay-synchronization policy layered on top of a continuous civil calendar. Version 2.0.0 promotes that content into the normative body of this standard: the ATF schedule and Minecraft time scale are now defined in Section II-B, and the conversion procedure built on it is defined in Section VII. The BIASR website’s Minecraft World Time panel and Official UBC Date readout now display the same underlying accumulated-day count (the BJDN of Section VII) in two equivalent framings, per Appendix B. Implementations persist the current civil day number and in-day clock tick only as a diagnostic record; both are fully recomputed from the Founding Epoch on every load (Section II-B), so a page refresh always restores the identical, deterministic state with no drift.

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- [4] Chief Scientific Committee on Temporal Standards, “CSCTS Report 00: Reconstruction of the Founding Epoch,” *Binodian Journal of Astronomical Science*, vol. 1, no. 1, pp. 1–14, UBC 1.
- [5] Chief Scientific Committee on Temporal Standards, “CSCTS Report 03: Measurement of the Binodian Tropical Year,” *Binodian Journal of Astronomical Science*, vol. 3, no. 2, pp. 45–62, UBC 3.

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